

LAB PROGRAMS (1-5)
COMPUTER VISION-ITA-0505
SLOT-D

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QUESTION-1

Perform basic Image Handling and processing operations on the image is to read an image in python and Convert an Image to Gray- scale.

PYTHON CODE:

```
import cv2

# Read the image
image = cv2.imread("sample.jpg")

# Check if the image is loaded
if image is None:
    print("Error: Image not found!")
    exit()

# Convert image to grayscale
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Display original image
cv2.imshow("Original Image", image)

# Display grayscale image
cv2.imshow("Gray-scale Image", gray_image)

# Wait for a key press
```

```
cv2.waitKey(0)
# Close all windows
cv2.destroyAllWindows()
```

OUTPUT:



QUESTION-2:

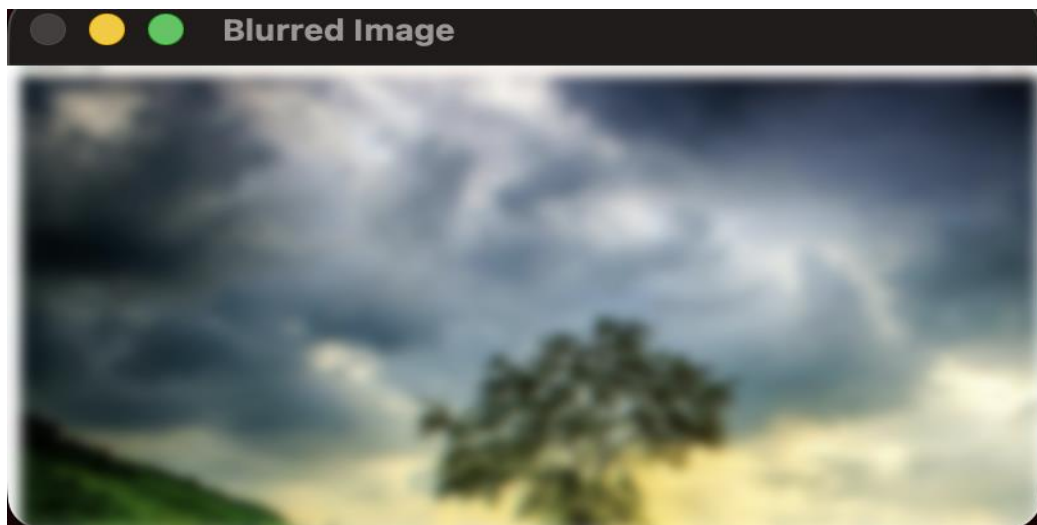
Perform basic Image Handling and processing operations on the image is to read an image in python and Convert an Image to Blur using Gaussian Blur

PYTHON CODE:

```
import cv2
# Read the image
image = cv2.imread("sample.jpg")
# Check if the image is loaded
if image is None:
    print("Error: Image not found!")
    exit()
# Apply Gaussian Blur
```

```
blur_image = cv2.GaussianBlur(image, (15, 15), 0)
# Display original image
cv2.imshow("Original Image", image)
# Display blurred image
cv2.imshow("Gaussian Blur Image", blur_image)
# Wait until a key is pressed
cv2.waitKey(0)
# Close all windows
cv2.destroyAllWindows()
```

OUTPUT:



QUESTION-3:

Perform basic Image Handling and processing operations on the image is to read an image in python and Convert an Image to show outline using Canny function.

PYTHON CODE:

```
import cv2

# Read the image

image = cv2.imread("sample.jpg")

# Check if image is loaded

if image is None:

    print("Error: Image not found!")

    exit()

# Convert to grayscale

gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Apply Canny edge detection

edges = cv2.Canny(gray_image, 100, 200)

# Display images

cv2.imshow("Original Image", image)

cv2.imshow("Edge Detected Image", edges)

# Wait for key press

cv2.waitKey(0)

# Close all windows

cv2.destroyAllWindows()
```

OUTPUT:



QUESTION-4:

Implement histogram equalization on the given image and compare it with the original image using Open CV.

PYTHON CODE:

```
import cv2

# Read the image
image = cv2.imread("sample.jpg")

# Check if image is loaded
if image is None:
    print ("Error: Image not found!")
    exit ()

# Convert to grayscale
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Apply Histogram Equalization
```

```
equalized_image = cv2.equalizeHist(gray_image)
# Display images
cv2.imshow("Original Grayscale Image", gray_image)
cv2.imshow("Histogram Equalized Image", equalized_image)
# Wait until a key is pressed
cv2.waitKey(0)
# Close all windows
cv2.destroyAllWindows()
```

OUTPUT:



QUESTION-5:

Write a Python function to analyze the histogram of the given input image based on color levels

PYTHON CODE:

```
import cv2

import matplotlib.pyplot as plt

def analyze_histogram(image_path):

    # Read the image

    image = cv2.imread(image_path)

    # Check if image is loaded

    if image is None:

        print("Error: Image not found!")

        return

    # Display the original image

    cv2.imshow("Original Image", image)

    # BGR color channels

    colors = ('b', 'g', 'r')

    # Create histogram plot

    plt.figure(figsize=(8, 5))

    for i, color in enumerate(colors):

        histogram = cv2.calcHist([image], [i], None, [256], [0, 256])

        plt.plot(histogram, color=color, label=f"{color.upper()} Channel")

    plt.title("Color Histogram Analysis")

    plt.xlabel("Pixel Intensity")

    plt.ylabel("Number of Pixels")

    plt.xlim([0, 256])

    plt.legend()
```

```
# Show histogram  
plt.show()  
  
# Wait for key press  
cv2.waitKey(0)  
  
cv2.destroyAllWindows()  
  
# Call the function  
analyze_histogram("sample.jpg")
```

OUTPUT:

